

NIPUN AGARWAL

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EDUCATION

Columbia University in the City of New York

New York, NY

Master of Science in Computer Science (Machine Learning Track; GPA: 3.65/4.0)

Dec 2024

Courses: Analysis of Algorithms, Machine Learning, Large Scale Distributed Systems, Advanced Deep Learning, High Performance ML, Data Visualization, Natural Language Processing, Artificial Intelligence

Graduate Teaching Assistant: Computer Systems for Data Science; Graduate Research Assistant II: Columbia Build Lab

University of Mumbai

Mumbai, IN

Bachelor of Engineering in Computer Engineering with Honors in Data Science (GPA 9.8/10.0)

May 2023

Courses: Data Structures, Operating Systems, Database Management Systems, Data Warehousing & Mining, Social Networks

TECHNICAL SKILLS

Programming languages: Python, Java, Golang, Ruby, C/C++, C#, YAML, JavaScript, PHP, SQL, R, Matlab

Libraries: PyTorch, OpenCV, NumPy, Pandas, Scikit-Learn, Matplotlib, TensorFlow, Keras, spaCy, LangChain, HuggingFace

Frameworks: Django, Flask, JUnit, React.js, Node.js, Hadoop, Bootstrap, Angular

Tools: AWS, Google Cloud Services, Docker, Kubernetes, Maven, Agile, Git, Jira, CI/CD, HTML, CSS, YAML

PROFESSIONAL EXPERIENCE

IBM – Columbia University in the City of New York

New York, NY

Research Collaborator

Oct 2024 – Dec 2024

- Refactored IBM's foundation models repository, integrating **DTensor APIs** to **optimize multi-GPU concurrency**, boosting **system throughput by 20%** across large foundation models (**LLaMA-7B, GPT, Granite-8B**).
- Introduced sequence parallelism techniques, **reducing training & inference memory overhead by 25%**, benchmarking performance improvements across diverse models.
- Collaborated with IBM research scientist Dr. Antoni Viros i Martin to optimize distributed deep learning workflows, ensuring **efficient parallelism** and **reduced compute overhead** in production-scale models.

Amazon Web Services (AWS) - SageMaker GroundTruth

Seattle, WA

Software Development Engineer Intern

Jun 2024 – Aug 2024

- Launched an automated mechanism that channels SageMaker performance data into both regional and enterprise-level S3 storage using **AWS Lambda, Java, YAML, Jinja, & SQL**. This change reduced **over 30+ hours** of manual effort each week while expanding data availability.
- Enabled immediate oversight and **detailed analytics** by linking the data flow to **Amazon Athena & AWS QuickSight**, advancing data presentation and accelerating strategic evaluations.
- Applied **prompt-based fine-tuning** to refine a **large language model** which turns **user-friendly prompts** into **SQL queries**, eliminating technical barriers for data retrieval and streamlining insights with AI-driven workflows.

Dream11 – World's Biggest Fantasy Sports Platform

Mumbai, IN

Software Development Engineer Intern

Aug 2022 – Sep 2022

- Built an IT Asset Management System employing **Python, Django & SQLite3 database** to gather and report data about company's infrastructure, resulting in **usage by 50+ team members**.
- Devised a program using **Python & Selenium** to automate the dissemination of **more than 50 reports daily** in PDF format on WhatsApp, **saving 5+ work hours every week**.

SOFTWARE DEVELOPMENT PROJECTS

Home Rental Platform | React, Node, Express, MongoDB, Tailwind

- Engineered a full-stack accommodation rental platform with a **React & Tailwind frontend** and a **Node.js, Express, MongoDB backend**, enabling booking, cancellation, and payments with a robust **RESTful API**.
- Deployed **user authentication, listing creation, and booking management**, ensuring a seamless user experience across devices with responsive UI and optimized backend workflows.

Model Checking Paxos | Go, MapReduce

- Developed a **scalable MapReduce framework** in **Go** with built-in word count, dynamic job distribution, and resilient worker failure handling for fault tolerance in distributed systems.
- Architected a **Paxos-based key/value store** in **Go**, addressing network and server challenges to maintain data consistency, and implemented a **model checker** verifying correctness in **diverse consensus scenarios**.

Summarizing Team Meetings | Python, Django, BART, Autoencoders, Transformers, TensorFlow

- Spearheaded a **Django-based REST API** for ingesting raw meeting transcripts & returning concise summaries, reducing the need for manual note-taking.
- Integrated **BART & Transformers** to compress lengthy discussions into short, actionable insights, handling parallel requests through concurrency controls in **Python** for **scalable performance**.
- Designed a modular architecture that parses data, applies text preprocessing, & delivers real-time summaries via JSON endpoints, **achieving 90% comprehension** based on user feedback.